Chapter 7. The Bootstrap

Sampling distributions, bootstrap estimates of standard error and bias, resampling schemes

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7.1 The plug-in principle

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $\hat{\theta} = T(X_1, \dots, X_n)$ estimate $\theta = \theta(F)$. The **sampling distribution** G_F of $\hat{\theta}$, and with it the bias, standard error and quantiles, depends on the unknown F. If we could draw new samples from F we would simply simulate. The bootstrap replaces F by an estimate $\hat{F}$ and simulates from $\hat{F}$:

$$\text{real world: } F \to \mathbb{X}_n \to \hat{\theta}, \qquad \text{bootstrap world: } \hat{F} \to \mathbb{X}_n^* \to \hat{\theta}^*.$$

The distribution of $\hat{\theta}^*$ given the data, $G^* = G_{\hat{F}}$, is the **bootstrap distribution**; it estimates G_F.

Nonparametric bootstrap. $\hat{F} = F_n$, the empirical distribution (mass $1/n$ at each observation). Sampling from F_n means drawing n observations *with replacement* from the data.

1. Draw a bootstrap sample $X_1^*, \dots, X_n^*$ with replacement from $\{X_1, \dots, X_n\}$ and compute $\hat{\theta}^* = T(X_1^*, \dots, X_n^*)$.
2. Repeat B times to get $\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}$; their empirical distribution approximates G^*.

There are two errors: $F_n \neq F$ (statistical, of order $n^{-1/2}$, the important one) and $B < \infty$ (Monte Carlo, of order $B^{-1/2}$, reducible at will).

7.2 Bootstrap estimates of standard error and bias

Replacing F by F_n in $\text{se}_F(\hat{\theta})$ and $\text{bias}_F(\hat{\theta}) = E_F\hat{\theta} - \theta(F)$ gives

$$\widehat{\text{se}}_{\text{Boot}} = \Big(\frac{1}{B-1}\sum_b (\hat{\theta}^{*b} - \bar{\theta}^*)^2\Big)^{1/2}, \qquad \widehat{\text{bias}}_{\text{Boot}} = \bar{\theta}^* - \hat{\theta}, \qquad \bar{\theta}^* = \frac{1}{B}\sum_b \hat{\theta}^{*b}.$$

Note that the role of the true $\theta(F)$ is played by $\theta(F_n)$, which is $\hat{\theta}$ for plug-in statistics.

Example 7.1 (An exact bootstrap distribution)

Data $\{1, 4, 9\}$, $\hat{\theta}$ = median = 4. Among the $3^3 = 27$ equally likely resamples, the median equals the smallest value 1 iff at least two draws are 1: probability $(3 \cdot 2 + 1)/27 = 7/27$. Likewise it equals 9 with probability 7/27, and 4 with probability 13/27. Hence

$$E^*\hat{\theta}^* = \frac{7 \cdot 1 + 13 \cdot 4 + 7 \cdot 9}{27} = 4.52, \qquad \text{se}^* = \Big(\frac{7 + 13 \cdot 16 + 7 \cdot 81}{27} - 4.52^2\Big)^{1/2} = 2.92,$$

and the bootstrap bias estimate is 0.52. With B random resamples these values are only approximated. Here they can be computed exactly because n is tiny.

When the bootstrap works

For smooth statistics (means, smooth functions of means, many M- and U-statistics) the bootstrap distribution of $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ consistently estimates that of $\sqrt{n}(\hat{\theta} - \theta)$. It can fail for non-smooth targets at the boundary of the parameter space, e.g. $\hat{\theta} = \max_i X_i$ for Uniform$(0, \theta)$, where $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$, whereas $P(\hat{\theta} = \theta) = 0$.

7.3 Parametric bootstrap

If $F = F_\eta$ belongs to a parametric family, estimate $\hat{\eta}$ (e.g. by maximum likelihood) and draw bootstrap samples from $F_{\hat{\eta}}$ instead of F_n. When the model is right this is more accurate than the nonparametric bootstrap, and it is useful when the sampling distribution of $\hat{\theta}$ is analytically intractable (e.g. a nonlinear functional of a Weibull or gamma fit). When the model is wrong it can be badly biased.

7.4 Resampling designs

The guiding rule is: *resample the way the data were generated.*

- **Several samples.** For independent samples from $F_1, \ldots, F_k$, resample within each sample separately, keeping the sample sizes $n_1, \ldots, n_k$.
- **Bivariate data, random design.** If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$, resample the pairs (**case** or pairs bootstrap). This keeps the dependence between X and Y and is valid even if a regression model is wrong.
- **Regression, fixed design.** If $Y_i = \alpha + \beta x_i + \epsilon_i$ with fixed x_i and i.i.d. errors, resample residuals: $Y_i^* = \hat{\alpha} + \hat{\beta} x_i + r_i^*$, with r_i^* drawn from the (leverage-adjusted) residuals. This is more efficient when the model is right but relies on it, including homoscedasticity.

Choosing between the two regression schemes:

- Classical regression inference is conditional on the x's. The residual bootstrap keeps the x's fixed, so it reproduces this setting and agrees closely with normal theory when that theory is adequate.
- The case bootstrap remains valid when the linear model is wrong. Then β is reinterpreted as the best linear approximation, the minimizer of $E(Y - \alpha - \beta X)^2$, and heteroscedasticity is allowed. The cost is some efficiency when the model is right.
- In designed experiments the x's are chosen by the experimenter, which argues for keeping them fixed.

Contrast with permutation tests: the bootstrap samples *with* replacement and keeps the structure of the alternative (e.g. the pairs (X_i, Y_i) stay together). A permutation test samples *without* replacement and deliberately destroys the association, to produce the null distribution.

7.5 Selected exercises

Motivation. When the statistic is simple, the ideal bootstrap ($B = \infty$) can be computed by hand. The exercises do this for the empirical distribution function, the mean and the variance, prove the failure for the uniform maximum stated above, and count bootstrap samples.

Exercise 7.1 (The empirical distribution function)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ and $F_n(x) = n^{-1} \sum_i I(X_i \le x)$. For x with $F(x) \in (0, 1)$, show:
(a) $nF_n(x) \sim B(n, F(x))$;
(b) $F_n(x)$ is unbiased for $F(x)$ with standard error $\sqrt{F(x)(1 - F(x))/n}$.

Solution.

(a) The indicators $I(X_i \le x)$ are independent, being functions of independent variables, and Bernoulli with success probability $P(X_i \le x) = F(x)$. Their sum $nF_n(x)$ is therefore $B(n, F(x))$.

(b) From (a), $E\, nF_n(x) = nF(x)$ and $\operatorname{Var} nF_n(x) = nF(x)(1-F(x))$. Divide by n and n^2.

So F_n, the distribution the nonparametric bootstrap samples from, is within $O_P(n^{-1/2})$ of F at each x. This is the statistical error of order $n^{-1/2}$ mentioned in Section 7.1. ■

Exercise 7.5 (Ideal bootstrap for the mean and the variance)

Let $x_1, \dots, x_n$ be fixed data with mean $\bar{x}$, $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$ and $s^2 = \frac{n}{n-1}\hat{\sigma}^2$, and let $X_1^*, \dots, X_n^*$ be drawn with replacement.
(a) Show that $E^*\bar{X}^* = \bar{x}$ and $\operatorname{Var}^* \bar{X}^* = \hat{\sigma}^2/n$.
(b) For $\theta = \mu^2$ and $\hat{\theta}_1 = \bar{X}^2$, show that the ideal bootstrap bias estimate is $\hat{\sigma}^2/n$. Compare it with the true bias, and find the bias of the corrected estimator $2\hat{\theta}_1 - \bar{\theta}^*$.
(c) For $\theta = \sigma^2$ and $\hat{\theta}_2 = s^2$, show that $\bar{\theta}^* - \hat{\theta}_2 = -s^2/n$, although s^2 is unbiased. Explain.

Solution [Supplement].

(a) The X_i^* are i.i.d. from F_n, with $E^*X_1^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\operatorname{Var}^* X_1^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$. Average n independent copies.

(b) By (a), $E^*(\bar{X}^{*2}) = \operatorname{Var}^* \bar{X}^* + (E^*\bar{X}^*)^2 = \hat{\sigma}^2/n + \bar{x}^2$. So $\bar{\theta}^* - \hat{\theta}_1 = \hat{\sigma}^2/n$. The true bias is $E\bar{X}^2 - \mu^2 = \operatorname{Var}\bar{X} = \sigma^2/n$; the bootstrap replaces σ^2 by $\hat{\sigma}^2$. The corrected estimator is $\bar{X}^2 - \hat{\sigma}^2/n$, and since $E\hat{\sigma}^2 = \frac{n-1}{n}\sigma^2$,

$$E(\bar{X}^2 - \hat{\sigma}^2/n) = \mu^2 + \frac{\sigma^2}{n} - \frac{(n-1)\sigma^2}{n^2} = \mu^2 + \frac{\sigma^2}{n^2}.$$

The bias drops from order n^{-1} to order n^{-2}.

(c) s^{*2} is unbiased for the variance of the distribution it is sampled from, here F_n with variance $\hat{\sigma}^2$. So $\bar{\theta}^* = E^*s^{*2} = \hat{\sigma}^2$ and $\bar{\theta}^* - s^2 = \hat{\sigma}^2 - s^2 = -s^2/n$. The bootstrap estimates $E_F\hat{\theta} - \theta(F)$ by $E_{F_n}\hat{\theta}^* - \theta(F_n)$, and here $\theta(F_n) = \hat{\sigma}^2$, not s^2. With the correct centering the estimate is $\hat{\sigma}^2 - \hat{\sigma}^2 = 0$. The shortcut $\bar{\theta}^* - \hat{\theta}$ is valid only for plug-in statistics, as remarked in Section 7.2. ■

Exercise 7.6 (Failure for the uniform maximum)

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$ and $\hat{\theta} = \max_i X_i$.
(a) Show that $n(\theta - \hat{\theta})/\theta \xrightarrow{d} \text{Exp}(1)$.
(b) Show that $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1}$.
(c) Conclude that the bootstrap distribution of $n(\hat{\theta} - \hat{\theta}^*)/\hat{\theta}$ does not converge to Exp(1).

Solution [Supplement].

(a) For $t \ge 0$ and $n > t$, $P(n(\theta - \hat{\theta})/\theta > t) = P(\hat{\theta} < \theta(1 - t/n)) = (1 - t/n)^n \to e^{-t}$.

(b) A.s. the maximum $x_{(n)}$ is a single data point. $\hat{\theta}^* = \hat{\theta}$ iff it is drawn at least once in n independent draws, each hitting it with probability $1/n$. So $P^*(\hat{\theta}^* = \hat{\theta}) = 1 - (1 - 1/n)^n \to 1 - e^{-1} \approx 0.632$.

(c) Let $V^* = n(\hat{\theta} - \hat{\theta}^*)/\hat{\theta} \ge 0$. If its bootstrap law converged to Exp(1), whose distribution function is continuous, then $P^*(V^* \le \frac{1}{2}) \to 1 - e^{-1/2} = 0.393$. But $P^*(V^* \le \frac{1}{2}) \ge P^*(V^* = 0) \to 0.632$ for every data set, a contradiction. The bootstrap law keeps an atom of mass about 0.632 at 0, while the target is continuous. The cause is that $\hat{\theta}$ sits at the boundary of the support, so F_n cannot mimic the behavior of F near θ. ■

Exercise 7.7 (Counting bootstrap samples)

Let the data $x_1, \ldots, x_n$ be distinct.
(a) Show that there are $\binom{2n-1}{n}$ distinct bootstrap samples (as multisets), and that the one in which x_i appears c_i times has probability $\frac{n!}{c_1! \cdots c_n!} n^{-n}$.
(b) Show that the expected number of distinct original observations in a bootstrap sample is $n\{1 - (1 - 1/n)^n\} \approx 0.632\, n$.

Solution [Supplement].

(a) A multiset is determined by the counts $(c_1, \ldots, c_n)$ with $c_i \geq 0$ and $\sum c_i = n$. Represent it by n stars and $n - 1$ bars, which gives $\binom{2n-1}{n}$ arrangements. The n^n ordered draws are equally likely, and the number of ordered draws with given counts is the multinomial coefficient $n!/(c_1! \cdots c_n!)$.

(b) Let $J_i = I(x_i \text{ appears})$. Then $EJ_i = 1 - (1 - 1/n)^n$, and the expected number is $\sum_i EJ_i = n\{1 - (1 - 1/n)^n\}$. Since $(1 - 1/n)^n \to e^{-1}$, this is about $0.632\, n$.

For $n = 3$ there are $\binom{5}{3} = 10$ multisets among the 27 ordered draws of Example 7.1, e.g. $\{1, 4, 9\}$ has probability $6/27$ and $\{1, 1, 1\}$ probability $1/27$. Exhaustive enumeration grows like $\binom{2n-1}{n} = \frac{1}{2}\binom{2n}{n} \approx 4^n/(2\sqrt{\pi n})$, which is why Monte Carlo resampling is used. ■